K-Means Clustering Fundamentals: Verifying the Algorithm Against Known Ground Truth

Clustering is unusual among machine learning tasks: there are no labels to score against, so on real data it is genuinely hard to know whether an algorithm's output is meaningful. This project sidesteps that problem deliberately by testing K-Means on synthetic data with known structure — 1,000 two-dimensional points generated in exactly five clusters using scikit-learn's make_blobs with a fixed random seed — so the algorithm's output can be verified against ground truth rather than taken on faith.

The workflow is intentionally minimal and transparent. K-Means is configured with five clusters and a fixed random_state of 42 (reproducibility matters here, since K-Means outcomes depend on centroid initialisation), fitted to the 1,000 points, and its two key outputs extracted: the label assignment for every point and the coordinates of the five learned centroids. The recovered centroids — located at approximately (−6.69, −6.81), (2.04, 4.27), (−2.50, 9.04), (−8.81, 7.40), and (4.67, 1.87) in the feature space — sit at the visual centres of the five generated blobs, and a colour-coded scatter plot with the centroids overlaid as red crosses confirms clean, well-separated cluster recovery.

Beneath the simplicity, the demonstration encodes the core mechanics of centroid-based partitioning: K-Means alternates between assigning each point to its nearest centroid and relocating each centroid to the mean of its assigned points, iterating until assignments stabilise. Watching the algorithm recover a known five-cluster structure builds the intuition needed to judge its behaviour later on real data, where the truth is unknowable.

The project is also honest about its boundary. Here the number of clusters, $k = 5$, was known in advance because the data was generated that way; on real-world data it never is. Selecting k requires the elbow method — plotting within-cluster sum of squares against candidate values of k — and silhouette analysis to score cluster cohesion and separation. These techniques, together with application to a real customer-segmentation dataset, are the stated next steps that will extend this fundamentals demonstration into a full unsupervised learning case study.

As it stands, the project delivers a clean, reproducible reference for how K-Means behaves when the answer is known — the necessary first step before trusting it when the answer is not.